Flexnetic: Cost-Effective and Smooth Evolution of Optical Backbone

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***Abstract*—The increasing traffic on WANs due to growing number of applications imposes significant strain on the infrastructure of cloud service providers. The primary expense in augmenting network capacity entails high costs associated with procuring expensive transponders that facilitate inter-regional optical signal transmission. The advent of spacing variable transponders facilitates cost-effective network upgrades and increased transmission efficiency, however, implementing such upgrades in one step may incur high costs and service disruptions. Achieving a cost-effective and smooth network upgrade presents considerable challenges in identifying critical IP links and minimizing costs within existing architecture. We introduce Flexnetic, a planning tool which utilizes a hybrid approach of both modern and legacy transponders, along with establishment of optical bypass, to accommodate the escalating traffic demands while minimizing the costs during network upgrades. Flexnetic incorporates two novel algorithms: a NLP model to maximize IP link capacity utilization, and an MIP model for efficient IP layer implementation at optical layer, emphasizing reuse of existing transponder and minimizes new device requirement. Our simulation of upgrade plans on common WAN topologies revealed superior performance, with up to 91.9% cost savings and $2.33\times$ capacity increase over existing state-of-the-art solutions, highlighting Flexnetic's potential for cost-efficient and capacity-optimized network upgrades.**

***Index Terms*—network planning, network management, optical backbone networks, network optimization, network modeling**

I. Introduction

Cloud service providers like Google, Microsoft, and Tencent manage global optical Wide Area Networks (WANs) that connect data centers and users. These networks handle traffic from online applications, like meetings and streaming, which often require real-time responses due to periodic or bursty patterns. Additionally, data transfers between data centers, such as big-data analytics, are usually scheduled during off-peak hours to utilize unused network capacity. The growing number of applications and demand for high-quality multimedia necessitate upgrades to these optical networks. Currently, WANs are connected by fibers and optical devices, forming a large-scale backbone. Capacity upgrades typically involve adding more transmission media (like long-distance fibers) or updating optical equipment. Installing more fibers involves significant construction, constrained by geography and policies, and incurs high costs, but is the most direct way to boost capacity. Optical devices in data center requires advanced implementations to increase capacity despite existing fiber and physical limits. Transponders convert electrical signals to optical signals (or vice versa) and provide transmission capacity. A typical legacy transponder pair offers 100Gbps capacity [1]. To meet capacity demands, purchasing numerous costly transponders is a significant upgrade expense. Recent studies [2] propose re-planning existing WANs by establishing optical bypasses, reducing optical-electrical-optical (OEO) conversions, and decreasing the number of required transponders. These transponders can be repurposed to enhance IP link capacity, offering additional capacity without extra costs.

Recent advances in optical technology have enabled high-bandwidth transponders to deliver up to 800Gbps within the same spectrum by improving spectral efficiency, making capacity upgrades more cost-effective. Recent research in FlexWAN [3] introduces a network planner with modern transponders. However, its approach of simultaneously upgrading all transponders is impractically expensive and may disrupt the network and violate Service Level Agreements (SLAs) due to high costs and interruptions during the upgrade.

An effective strategy to address link capacity shortages is to upgrade specific links in the optical network with modern transponders. By deploying these devices on critical links and keeping legacy devices elsewhere, the network can be efficiently enhanced. Upgrading only bottleneck links minimizes traffic disruption and supports network evolution. However, determining which IP links to upgrade and procuring modern transponders at the lowest global cost remain challenges. First, identifying potential bottleneck links is crucial. Unlike traditional TE which focuses on managing existing capacity by balancing traffic load. We use industry methods to distribute traffic across the nearest k links to prevent overload. Second, mapping IP links to the optical spectrum on specific fibers must satisfy various constraints, making formulation and solution challenging. Finally, optimizing upgrade costs and network flexibility based on fiber topology involves trade-offs. Aggressive strategies like optical bypass reduce transponder costs but limit flexibility, while conservative approachs may not adequately control upgrade costs.

To address these issues, we introduce Flexnetic, a planning tool for cloud service operators to seamlessly upgrade networks. It integrates with existing WAN optical backbones and

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traffic forecasts, which are typically well-known to administrators. Flexnetic optimizes the reuse of legacy transponders and employs strategies like creating optical bypasses and acquiring high-bandwidth transponders. This approach minimizes the need for new optical devices while ensuring the WAN can effectively accommodate growing demands. Flexnetic tackles two key challenges: accurately estimating required IP link capacity within the current WAN topology and traffic matrix, and implementing it at optical layer.

To address the first challenge, we utilized a nonlinear programming (NLP) model during the capacity prediction stage to allocate traffic demand on the network. This model aims to minimize the temporal variation in traffic required by each IP link, thereby maximizing the utilization of IP link capacity over time. This approach responsibly utilizes the transmission capabilities provided by the expensive transponders we purchased, minimizing waste of WAN capacity resources.

To address the second challenge, in the network planning stage, we introduced a mixed-integer programming (MIP) optimization model. This model is designed to realize generated IP network topology, which includes capacity demands, within the physical constraints of the optical network. It minimizes the number of transponders needed to implement the IP network and thoroughly considers reusing existing legacy transponders in this process.

We assessed Flexnetic's smooth network evolution upgrade plans on popular WAN topologies, noting up to 91.9% cost savings and a capacity increase of up to 2.33× compared to conventional optical bypass network planning. These impressive results highlight Flexnetic's potential for significant cost reduction and capacity enhancement in real-world network upgrades. Additionally, Flexnetic facilitates the transition from traditional fixed grid to flexible grid spectrum models, prioritizing budget allocation to the most critical high-traffic links. This makes it an appealing choice for cloud service providers.

In summary, we have made the following contributions:

- We develop a planning tool for the smooth evolution from existing Bandwidth Variable Transponders (BVT) optical networks to Spacing Variable Transponder (SVT) optical networks, which is designed to solve large-scale planning problems involving the allocation of optical spectrum and transponder configurations under specific constraints.
- We provide a comprehensive mathematical representation for establishment of optical bypasses and the hybrid utilization of existing BVTs and newly purchased SVTs.
- We propose a heuristic approach to balance the tradeoff between upgrade costs and network flexibility under limited network topology, which depends on the establishment of optical bypasses.
- We have released open-source code and computational components at https://github.com/flynnoct/flexnetic.

II. Background and Opportunities

A. Optical Bypass

In Figure 1, traffic travels through a fiber-based WAN backbone and enters a MUX/DEMUX port via Reconfigurable Optical Add-Drop Multiplexer (ROADM) devices. Mixed *colored* optical signals with distinct wavelengths are separated into *unicolor* signals, each matched to a specific modulation format and fed into router-mounted transponders. Each transponder converts the signal from optical to electrical format according to the modulation, and outputs it to the router. Therefore, routers process only electrical signals, abstracted from the optical links and spectrum allocations. In traditional Bandwidth Variable Transponder (BVT), the *unicolor* optical signal's spectrum width is fixed (e.g., 50GHz or 75GHz), forming fixed grid slots for signal processing. Inversely, advanced Spacing Variable Transponder (SVT) divide the spectrum into 12.5GHz pixels, with a single SVT occupying 4 (50GHz) to 12 (150GHz) pixels per modulation choice. This completes optical-electrical-optical (OEO) conversion and underscores the transponder's crucial role in the network.

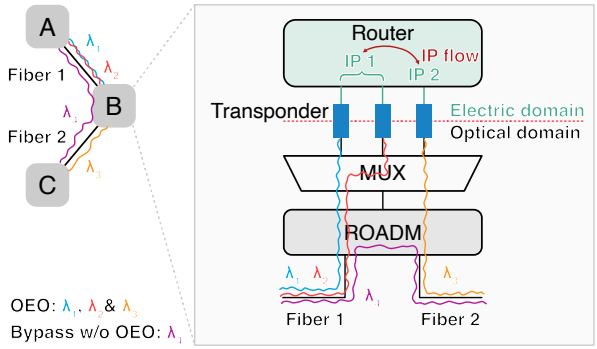

Fig. 1. OEO conversion and optical bypass.

In networks without optical bypass, an optical signal traversing a region undergoes full OEO conversion using two transponders: one converting the optical signal to electrical and another for the reverse. As the traffic is not destined for this region, the router simply forwards it, revealing an optimization opportunity. As shown in Figure 1, establishing an optical bypass (λ_4) avoids unnecessary OEO conversions, saving at least two transponders. The savings scale with the number of sub-links used by the IP link via link aggregation [4].

B. Spacing Variable Hardwares

The limited and costly spectrum resources on fibers result from the high expenses of laying and maintaining long-distance fibers. Research such as FlexWAN [3] explores the performance differences between BVT modulation formats and the Shannon Limit, introducing an improved optical device design, the SVT. SVTs can adjust parameters based on factors like bandwidth requirements, spectrum width, Optical Signal-to-Noise Ratio (OSNR), and transmission distance to optimize spectrum use. For example, an SVT can use a 50 GHz spectrum width with a 9.8 dB OSNR over 3000 km for 100 Gbps transmission. For 500 Gbps needs, SVTs can expand to a 137.5 GHz spectrum width, achieving a 19 dB OSNR for up to 1300 km of transmission.

Spacing Variable Hardware embodies the technological advancements essential for future optical infrastructures. By flexibly combining bandwidth, spectrum width, OSNR, and distances, these devices enable precise resource allocation and enhance spectrum utilization over diverse distances.

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C. Smooth Network Evolution

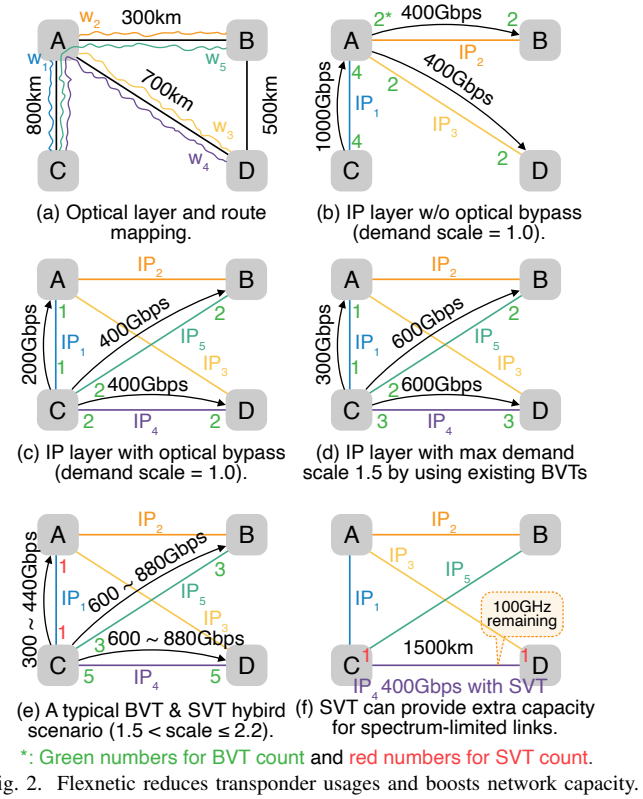
Real-world WANs contain numerous devices at different network levels, requiring gradual evolution as traffic demands grow. Due to the high cost of optical devices, a one-step upgrade is impractical and disruptive. Instead, a smooth upgrade approach, upgrading gradually to minimize waste, is preferred.

At our research time, major cloud providers primarily used BVTs with fixed spectrum width as described in Section II-A. SVTs, however, use spectrum more efficiently, enhancing capacity and flexibility. Although SVTs are ideal for upgrades, replacing all BVTs at once in large-scale WANs is too costly.

Fortunately, fiber spectrum properties allow for backward compatibility, permitting a mix of BVTs and SVTs without spectrum conflicts. This strategy increases capacity at bottlenecks using SVTs while keeping BVTs on resource-rich links to reduce costs. Thus, we created Flexnetic to aid network operators in smoothly evolving their WANs.

III. Motivation

Large cloud service providers manage global or national WANs. As services grow and quality demands increase, WAN traffic surges, often exceeding hardware capacities and necessitating upgrades. However, widespread hardware updates are costly and complex. A strategy of smooth network evolution involves incremental improvements to alleviate bottlenecks cost-effectively. Flexnetic leverages this by reusing optical devices, forming optical bypasses to utilize unused spectrum, increasing capacity without SVT purchases. Only when BVTs, even in optimal configurations, are still insufficient to meet demands, do we consider purchasing advanced SVT for bottleneck links, significantly reducing the WAN upgrade budget.



(a) Optical layer and route mapping.
(b) IP layer w/o optical bypass (demand scale = 1.0).
(c) IP layer with optical bypass (demand scale = 1.0).
(d) IP layer with max demand scale 1.5 by using existing BVTs
(e) A typical BVT & SVT hybird scenario (1.5 < scale ≤ 2.2).
(f) SVT can provide extra capacity for spectrum-limited links.

*: Green numbers for BVT count and red numbers for SVT count.

Fig. 2. Flexnetic reduces transponder usages and boosts network capacity.

A. Reduce Transponder Usage with Bypass

Consider an optical network topology shown in Figure 2(a), where nodes are connected by fibers with indicated lengths. At a certain time, traffic demand from Region C to Regions A, B, and D is 200Gbps, 400Gbps, and 400Gbps, respectively. In a traditional network using only BVTs without optical bypasses, the IP network topology is shown in Figure 2(b). This configuration requires a total of 1000Gbps between Region C and Region A, and 400Gbps between Regions A and B, and A and D. The BVT modulation formats indicate that 8, 4, and 4 BVT devices are needed for the A-C, A-B, and A-D links, respectively, totaling 16 devices. With Flexnetic, the IP network in Figure 2(c) uses optical bypasses between Regions B and C, and C and D, with traffic rerouted accordingly. In this setup, only 2, 4, and 4 BVT devices are required for the A-C, A-B, and A-D links, respectively. This adjustment saves 6 BVT devices while maintaining network capacity.

B. Enhanced Network Capacity Limits

Flexnetic enhances network capacity by optimizing transponder use in two key ways: first, by implementing optical bypasses to save BVTs, which can then be deployed on fibers with spare spectrum, providing higher capacity without additional costs. Second, advanced SVTs offer greater capacity within the same spectrum width compared to BVTs. In Figure 2(c), if traffic increases to 1.5 times the original volume, as in Figure 2(d), we can add a pair of BVTs to the C-D link. This requires 12 BVTs to meet the increased demand, still fewer than a traditional IP network would need. If traffic demand exceeds 1.5 times, existing transponders won't suffice. In such cases, Flexnetic introduces additional SVTs to boost capacity. The optimal solution involves integrating two SVTs to accommodate demand scaling up to 2.2 times. Figure 2(e) illustrates the network's optical topology with the added SVTs.

C. Establishing Further Optical Bypasses

The advanced modulation formats of SVTs significantly facilitate the construction of long-distance optical bypasses. Our earlier discussions assume ample unused spectrum resources on fibers, which is generally true, but in some cases, spectrum resources may be limited. In such situations, Flexnetic provides additional advantages for establishing optical bypasses. Consider the network topology in Figure 2(a), where there's a need to transmit 300Gbps between Regions C and D, but only 100GHz of the spectrum is available, as depicted in Figure 2(f). BVTs, constrained by distance, can only utilize 75GHz to create an optical bypass, supporting a maximum of 200Gbps—insufficient for the flow. This requires dismantling the bypass and rerouting traffic with two additional BVTs along the C-A-D path. However, SVTs can use the full 100GHz, providing 400Gbps capacity over 1550km, thus completing the transmission. In this instance, SVTs double the IP link capacity by using an extra 25GHz. Flexnetic aims to offer optimal IP link configurations, whether reducing spectrum resource usage or enhancing transmission capacity under limited spectrum conditions.

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D. Challenges

Intelligently Implementing Optical Bypasses in WANs. The complex topology of WANs means that indiscriminate placement of optical bypasses can negatively impact network capacity, availability, and flexibility. Constructing effective optical bypasses requires careful consideration of factors such as network topology, scale, and budget. Section IV-B explores this issue and introduces heuristic methods to guide the strategic placement of optical bypasses.

Accurately Predicting Traffic Demand in WANs. In WANs, inter-regional traffic generated by applications can be represented through matrices provided by cloud service providers, based on extensive observation and statistical analysis. The challenge lies in efficiently distributing this traffic across multiple IP links without wasting optical layer capacity, especially without knowing the IP link capacities. Section IV-C introduces an NLP model designed to balance traffic distribution across IP links effectively.

Solving Optical Layer Planning within Limited Time. We frame the optical planning problem as a MIP model. Solving this large-scale MIP on a vast WAN topology within a limited time is challenging. Section VI-C explores the process of solving MIP problems using Gurobi [5] and discusses the practical implications of accepting sub-optimal solutions to achieve feasibility.

IV. System Architecture

A. Overview

To optimize the use of existing BVTs and minimize the cost of acquiring SVTs during smooth evolution, we developed Flexnetic. This network planning tool aids cloud providers in upgrading WANs with SVTs. Flexnetic configures the IP layer topology and transponder settings in the optical layer to ensure sufficient capacity for the IP network. As depicted in Figure 3, Flexnetic's architecture and workflow begin with inputs of optical and existing IP network topologies. The pre-computation phase identifies potential optical bypass connections and important metadata, like the number of current BVTs. These bypasses help establish the upgraded WAN's IP topology, although IP link capacities are initially unassigned. Next, traffic demand matrices are fed into a capacity predictor, along with the IP topology. The predictor assigns suitable capacities to each IP link to meet regional traffic demands and inputs these into a planning solver. The planning solver tackles the problem as an enhanced RWA issue, using MIP within Gurobi solver [5]. It calculates the mapping between IP links and the transponders and spectrum in the optical layer, determining the transponder configuration scheme. These outcomes are provided for network operators to review or implement.

B. Strategies for Establishing Optical Bypasses

As previously outlined, Flexnetic establishes the IP link connection topology during the pre-computation phase, serving as input for subsequent modules. Consequently, devising a strategy for creating optical bypasses is the first step. The simplest method involves finding k shortest paths between region pairs.

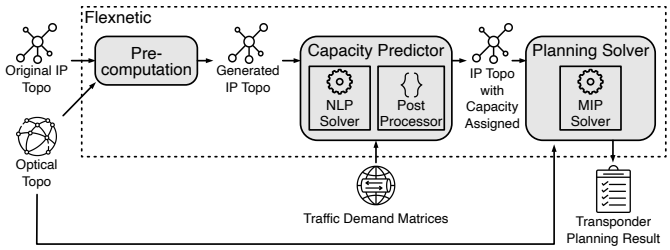

Fig. 3. System architecture of Flexnetic.

Traditional approaches focused mainly on SNR constraints due to transmission distance, a key factor limiting longer optical bypasses. However, we also recognize the importance of considering the network's optical topology when generating candidate optical bypasses. Overly aggressive bypass creation can compromise network flexibility and reduce spectrum efficiency, particularly on critical fibers. Therefore, balancing generation of more bypasses to minimize transponder usage against the need to maintain network simplicity and scalability has become a trade-off that needs to be handled in practice.

Low-connectivity Network Topologies and Optical Bypass Establishment. In bidirectional networks, a single connection that divides the network into two parts is a *bridge*. For instance, Figure 4(a) illustrates two Metropolitan Area Networks (MANs): MAN 1 (nodes A, B, C) and MAN 2 (nodes D, E, F, G), with the connection between C and D as a bridge. Cutting this link isolates MAN 1 from MAN 2. In this scenario, links like E-G, though also bridges, do not pose overload risks due to G being a single region. Bottlenecks frequently arise in WAN topologies where subnetworks connect at points with multiple regions. Using the KSP method for optical bypasses can intensify the load on bridges, exemplified by the creation of 11 bypasses on the C-D bridge for non-adjacent nodes ($\{A, B, C\} \times \{D, E, F, G\}, k = 1$). This leads to three primary concerns: requiring at least 550GHz of spectrum (50GHz $\times$ 11); potential resource wastage from reserved but underutilized spectrum; and the spectrum continuity constraint complicating allocation. To counter these, an algorithm begins with KSP, selecting the shortest k bypass candidates for each regional pair, then identifying fibers traversed by more candidates than a defined *bypass threshold* (h), marking such fibers as bottlenecks and eliminating their bypass candidates. In Figure 4(c), fibers 7 and 13 demonstrate up to 34 bypasses on a network with a 4.8THz spectrum, reaching a 35.4% occupancy. Balancing h minimizes upgrade costs while preserving network health, adaptable to topology and budget. As shown in Figure 4(b), selecting h ($1 < h < 11$) refines bypass configurations by removing those on critical bridges.

C. Capacity Predictor

In WAN productions, traffic demands between regions often change significantly over time. These varying demands are represented by matrices, each indicating the demand between regions at a specific time. Typically, each demand matrix is used to establish tunnels between regions, allocating traffic via traffic engineering or control strategies. As illustrated in Figure 3, Flexnetic's capacity predictor uses these demand matrices

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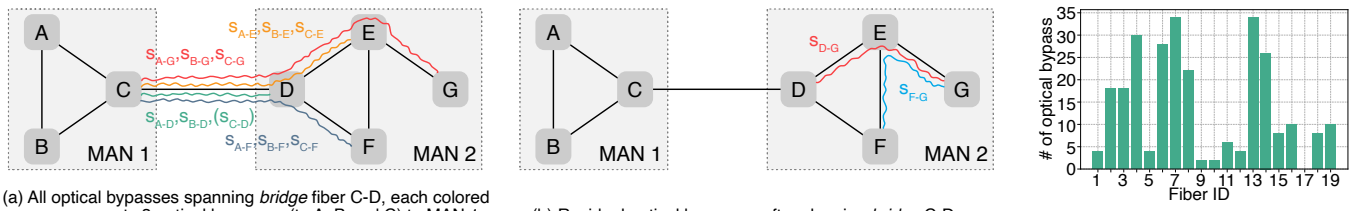

(a) All optical bypasses spanning *bridge* fiber C-D, each colored curve represents 3 optical bypasses (to A, B and C) to MAN 1, thus there are 12 optical bypasses on *bridge* C-D totally.

(b) Residual optical bypasses after cleaning *bridge* C-D up.

(c) Number of Optical Bypasses Carried by Each Fiber in B4 (k=1)

Fig. 4. Establishing optical bypasses on bridge linking two MANs risks overload.

to build tunnels using the KSP algorithm and allocate demand with a prediction algorithm. The predictor then assesses all tunnels on each IP link, accommodating all demand matrices with a comprehensive IP link capacity plan. Previous approaches, like Shoofly, chose multiple maximum traffic values for each region pair across different matrices, not concurrent in the network, which can be inefficient due to differences among matrices at different times. Using maximum values as predictions leads to significant capacity waste. Flexnetic uses a nonlinear programming model to distribute traffic across tunnels connecting regions, aligning capacity needs across matrices and minimizing variations in IP link capacity demands. In addition, Flexnetic preserves capacity margins during planning to handle sudden traffic spikes.

D. Planning Solver

With the pre-computed IP link topology and capacity plan, Flexnetic focuses on utilizing the optical layer to support the IP layer's transmission needs. It first prioritizes the use of existing BVTs. If the adjusted traffic demand can be met by modifying current devices, there's no need for cost-incurring SVTs. However, if a solution with only BVTs isn't feasible, Flexnetic addresses network bottlenecks by deploying BVTs on critical links, optimizing spectrum usage and bandwidth until the most cost-effective solution (minimum SVTs) is achieved. This process is governed by a large-scale MIP model, implemented via the Gurobi solver. Detailed mathematical formulations of this model are provided in Section V-B.

V. Formulation

As shown in Figure 3, Flexnetic employs a two-phase algorithm to address the distribution of demand matrices across network links, operating on different hierarchical layers. The first phase, capacity prediction, and the second phase, network planning, together use advanced SVTs and existing BVTs to devise an efficient arrangement strategy for these devices within a production WAN.

A. Capacity Prediction

The capacity prediction phase in Flexnetic allocates traffic flows, represented by traffic matrices, within a pre-built IP topology. A general-purpose capacity prediction model is implemented using nonlinear programming (NLP) on Gurobi [5] commercial solver. This model predicts the necessary capacity of IP links in typical scenarios, as detailed in Algorithm 1.

Capacity prediction inputs. Flexnetic constructs an initial IP topology graph, $G(V, E, S)$. Here, E contains simple IP links without optical bypass, and S includes optical bypass IP links. Collectively, E and S form the IP link set P, allowing $G(V, E, S)$ to be represented as $G(V, P)$. The set of demand matrices $\{D^m\}$ captures traffic demands across the network, with each D^m representing demands on the WAN at a specific moment. For each demand d_n^m, the corresponding tunnel set $T_{d_n^m}$ is established using the K-Shortest-Paths (KSP) method, identifying the shortest k paths between node pairs. The binary notation $L[t, p]$ indicates whether tunnel t traverses IP link p.

Algorithm 1 : Flexnetic Capacity Prediction

Inputs:

$G(V, P)$: Pre-computed IP topology.

$\{D^m\}$: A set of traffic demand matrices.

$d_n^m \in D^m$: The n-th demand value of D^m.

$T_{d_n^m}$: A set of tunnels designed to carry d_n^m.

$L[t, p]$: Equals 1 if tunnel t passes through IP link p, otherwise equals 0.

Decision Variables:

a_t: The size of flow passing through tunnel t.

c_p^m: Capacity of IP link p to meet traffic matrix D^m.

$$\textbf{Minimize}: \quad \sum_p \sum_m (c_p^m - \tfrac{1}{m} \cdot \sum_m c_p^m)^2 \qquad \text{(I)}$$

subject to:

$$\forall m, n: \quad \sum_{t \in T_{d_n^m}} a_t \geq d_n^m \qquad (1)$$

$$\forall m, p: \quad \sum_n \sum_{t \in T_{d_n^m}} (a_t \cdot L[t, p]) \leq c_p^m \qquad (2)$$

Post Processing:

$c_p = \max(c_p^m)$: Capacity of IP link p.

Capacity prediction decision variables. a_t notates the volume of flow allocated to tunnel t. Recall that a flow corresponds to a unique demand d. In other words, a_t represents the sub-flow allocated to a specific tunnel t to meet traffic demand. c_p denotes the capacity demand of each IP link and will be transmitted to network planning phase.

Capacity prediction objective function. (Expr. (I) in Alg. 1) Our objective function aims to optimize traffic distribution and network performance by minimizing periodic capacity demand fluctuations. The goal is to reduce the sum of variances in capacity required by all IP links across different traffic matrices. This approach minimizes global redundancy in IP link capacity by flexibly distributing traffic across tunnels, thereby improving the capacity utilization rate of IP links.

Flow allocation constraint. (Eq. (1) in Alg. 1) For each demand, the flow allocated across multiple tunnels must meet or exceed the demand value. Unlike some approaches that relax this constraint with additional variables, allowing tunnels

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to provide less than the required demand, we maintain this stricter constraint. This ensures a clear assessment of solution feasibility in the subsequent planning phase.

Flow capacity constraint. (Eq. (2) in Alg. 1) Tunnels for each demand are established over IP links, with the traffic allocated to them constrained by the IP link capacities. Each IP link must have a capacity greater than or equal to the total flow of all tunnels it carries simultaneously.

With these formulations, we have developed the IP link capacity prediction phase of Flexnetic. This algorithm initially assumes infinite IP link capacity, using only the IP topology to allocate traffic based on the objective function. It then records the capacity requirements of each IP link to meet the demand. The maximum capacity for each IP link across different traffic matrices is extracted for use in the network planning phase.

B. Network Planning with SVTs

The network planning phase in Flexnetic enhances the Routing and Wavelength Assignment (RWA) problem [6], targeting feasible wavelength programming on fiber spectrums. The use of SVTs boosts flexibility and performance but adds computational complexity. We have developed a MIP model for spectrum representation using pixel slots and implemented it with Gurobi. In practice, our algorithm performs efficiently, offering planning results that significantly reduce costs compared to traditional optical bypass or SVT planning methods.

Planning inputs. To represent the existing IP network, we use the current operational IP topology, denoted as $G'(V', E')$, as the starting point for planning. The optical network topology, which is described as graph $G_o(V_o, \phi)$, including both regions and fibers ϕ connecting these sites. l_p denotes the length of IP link (l_e and l_s). Note that for a IP link transmitted through a single fiber, its length corresponds to the length of the fiber. In contrast, an optical bypass link may span multiple fibers, in which case l_s is quantified as the cumulative length of the fibers it encompasses. $M[p, \phi]$ is a binary auxiliary notation indicating whether IP link p is constructed atop fiber ϕ. c_p is the output of the capacity prediction phase, representing the capacity demand of each IP link. The SVTs presents a set of modulation formats with varying spectrum widths, as previously discussed in Section II-B, each format occupies a spectrum width equal to an integer multiple of σ_j times width of a pixel (12.5 GHz). Given the objective phenomenon of a gradual SNR degradation over long-distance optical signal transmission, each format is assigned a maximum transmission distance ϵ_j and its corresponding maximum bandwidth capacity u_j achievable within such range. As shown in Table 2 of FlexWAN [3], BVTs' modulation formats can also be considered as special cases (employing widths of $75GHz = 6 \times 12.5GHz$). Hence, it is feasible to amalgamate the assorted formats into a singular modulation scheme table, streamlining the subsequent formulation of constraints. To differentiate the usable formats between SVT and BVT, an additional binary indicator β_j has been implemented to designate the applicability of a given format to SVT or BVT. j is utilized as the unique index for tabulating modulation formats, which means each value of j denotes a specific modulation format. For instance, $j = 4$ indicates that this pair of BVTs occupies a bandwidth of 62.5GHz on the spectrum, capable of providing a capacity of up to 200Gbps within a range of 1500km. We have compiled all the formats into a comprehensive list denoted by $[(\epsilon_j, \sigma_j, u_j, \beta_j)]$. $\delta^w_{j,q}$ represents an array of pre-computed indicators that denote whether the j-th modulation format, starting from the q-th pixel slot, will occupy the w-th slot. This can be readily computed through the modulation format table.

Planning decision variables. The pivotal insight in addressing SVT network planning issues with Flexnetic is how to modeling the problem comprehensively. Consequently, the establishment of decision variables to manage the boundaries of the problem is of paramount importance. These variables

Algorithm 2 : Flexnetic Network Planning

Inputs:

$G'(V', E')$: Currently operational IP network topology.

$G_o(V_o, \phi)$: Optical network topology.

l_p : Length of IP link p.

$M[p, \phi]$: Equals 1 if an IP link p is constructed based on fiber ϕ, otherwise equals 0.

c_p : Output of Algorithm 1, capacity demand of IP link p.

$[(\epsilon_j, \sigma_j, u_j, \beta_j)]$: A list of SVT modulation format properties including ϵ_j for signal maximum transmission reach, σ_j for the count of spectrum pixels occupied, u_j for maximum bandwidth capacity and β_j for indicating it's a BVT format (equals 1) or SVT format (equals 0).

$\delta^w_{j,q}$: Whether the j-th modulation format, starting from the q-th pixel slot, will occupy the w-th slot.

Decision Variables:

$\xi^p_{\phi,w}$: Equals 1 if a given pixel slot w on fiber ϕ is occupied by an IP link p, otherwise equals 0.

$\gamma^p_{j,q}$: Equals 1 if an IP link p utilizes the modulation format j starting at the q-th pixel slot, otherwise equals 0.

λ^p_j : Quantity of transponders utilized for an IP link p with modulation format j.

ρ^p : Equals 1 if IP link p uses BVTs, and equals 0 if uses SVTs.

$$\textbf{Minimize}: \sum_p \sum_j (1 - \rho^p)\lambda^p_j \tag{II}$$

subject to:

$$\forall p: \quad \sum_p \sum_j \rho^p \cdot \lambda^p_j \leq Z \tag{3}$$

$$\forall p, q, j: \quad \gamma^p_{j,q} \cdot (\rho^p - \beta_j) = 0 \tag{4}$$

$$\forall \phi, w: \quad \sum_p \xi^p_{\phi,w} \leq 1 \tag{5}$$

$$\forall s, w: \quad \xi^s_{\phi,w} = \xi^s_{\phi',w}\ , \text{ if } M[s,\phi] = M[s,\phi'] = 1 \tag{6}$$

$$\forall p, j: \quad \lambda^p_j \geq 0 \tag{7}$$

$$\forall p, j: \quad \lambda^p_j \cdot (u_j - l_p) \geq 0 \tag{8}$$

$$\forall p: \quad \sum_j \lambda^p_j \cdot u_j \geq c_p \tag{9}$$

$$\forall \phi: \quad \xi^p_{\phi,w} = (\sum_j \sum_q \gamma^p_{j,q} \delta^w_{j,q}) \cdot M[p,\phi] \tag{10}$$

$$\forall p, j: \quad \sum_q \gamma^p_{j,q} = \lambda^p_j \tag{11}$$

$$\forall p, q, j: \quad \gamma^p_{j,q} = \gamma^{p'}_{j,q}\ , \text{ if } p' = p.\text{reversed} \tag{12}$$

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are capable of explicitly representing the state of each pixel on each fiber. We subdivided the spectrum of each optical fiber into an array of pixel slots indexed by w, and denoted by binary $\xi^p_{\phi,w}$ the occupancy of a given pixel slot w by an IP link p traversing fiber ϕ. Subsequently, we employed the variable q to signify the start slot index of the modulation format, with the binary variable $\gamma^p_{j,q}$ indicating whether an IP link utilizes the modulation format (indexed by j) starting at the q-th pixel slot. Note that while the total counts of q and w both equate to the sum of pixel slots on a fiber, their respective implications are distinctly different. For instance, when constructing optical bypass s_0 on fiber ϕ_0 and selecting a 50GHz modulation format $j = 2$ that occupies pixel slots 4 through 7 on ϕ_0, we would have $\gamma^{s_0}_{2,4} = 1$ and $\gamma^{s_0}_{2,5} = \gamma^{s_0}_{2,6} = \gamma^{s_0}_{2,6} = 0$, whereas $\xi^{s_0}_{\phi_0,4} = \xi^{s_0}_{\phi_0,5} = \xi^{s_0}_{\phi_0,6} = \xi^{s_0}_{\phi_0,7} = 1$. This denotes that while the modulation format begins at the fourth pixel slot, it does not commence anew at subsequent slots within its allocation, whereas the IP link is present in each of the occupied slots from the fourth through the seventh. Link aggregation [4] enables the employment of multiple parallel transponders to build a single IP link, thereby augmenting capacity. The symbol λ^p_j denotes the quantity of transponders utilized for an IP link, which are specified by the j-th type of modulation format. Binary variable ρ^p signifies the selection of either SVTs or BVTs for a given IP link p.

Planning objective function. (Expr. (II) in Alg. 2) Flexnetic aims on reducing the overall cost during WAN upgrade, which can be described as minimizing the quantity of newly purchased SVTs while making full use of legacy BVTs. BVTs are existing devices in WAN which means reuse these devices will not incur additional costs and will be filtered out by $1 - \rho^p$. Hence, the primary costs associated with the upgrade are directly proportional to the quantity of SVTs required to be purchased which can be described as this formula.

Fix transponder usage constraint. (Eq. (3) in Alg. 2) Flexnetic strives to maximize the reuse of BVTs in order to minimize the necessity for purchasing new SVTs. This approach is one of the primary measures for reducing costs during the gradual evolution of the network. Z is a fixed value derived from the input IP network topology $G(V, E, S)$, representing the number of BVTs used in the existing IP network. The constraints ensure that Flexnetic does not exceed the original quantity of BVTs in its newly planned network.

Modulation format constraint. (Eq. (4) in Alg. 2) Flexnetic selects appropriate modulation formats for each IP link, with each format being specifically compatible with either BVTs or SVTs. For the sake of operational feasibility in configuration process, Flexnetic prohibits the use of multiple groups of transponders from spanning different types on same IP link, which means an IP link must be constructed either entirely from SVTs or entirely from BVTs, without any mix of the two types being permitted.

In this section, we discuss the mathematical model used in the Network Planning phase of building Flexnetic. For the optical layer constraints of SVTs (Eq. 5 - 12 in Alg.2), FlexWAN [3] has already provided a comprehensive introduction.

VI. Evaluation

We implemented Flexnetic's optimization algorithm to assess its benefits and cost savings in real-world WAN evolution. Benchmarks show that solving all topologies takes only a few hours, suitable for an offline network planner.

Optical Specifications. Optical devices from various manufacturers can differ in specifications and performance. To address this, we standardized the optical device specifications for our benchmarks. During evaluation, all BVTs used a fixed 75GHz spectrum spacing, offering the distance reach and transmission bandwidth listed in Table 2 of Miao et al. [3].

Benchmark schemes. Flexnetic employs optical bypass and advanced SVTs to optimize costs during network upgrades. We evaluated its performance on real WAN topologies, comparing it with schemes like Shoofly-improved and FlexWAN, which focus on minimizing transponder usage.

- **Shoofly-improved.**Shoofly, a recently proposed method, constructs optical bypasses using only BVTs to maximize transponder savings. For benchmark alignment, we configured Shoofly to reuse existing BVTs and add more as needed during capacity shortages. Additionally, we reinforced previously ignored optical layer constraints and removed the 3-hop limit on bypasses in Shoofly. We term this enhanced version Shoofly-improved.
- **FlexWAN.** FlexWAN initially demonstrated the use of SVTs in WANs and provided a detailed optical layer description. Its planning algorithm minimizes SVT usage and serves as a tool for designing advanced WANs built exclusively on SVTs from the ground up.

Topology and Demands. We evaluated four WAN topologies: two private (Experimental and TWAN) and two public (CERNET [7] and B4). From IP links, we derived the initial number of BVTs in the optical networks. Utilizing the long-tail distribution from SMORE [8], we generated multiple demand matrices for each topology to reflect typical periodic variations in real traffic. We further assessed capacity and performance by adjusting the demand matrices with a scale factor.

A. Cost Saving with Flexnetic

Flexnetic aims to minimize the need for new transponders during network upgrades. Our simulation results in Figure 5 demonstrate that Flexnetic significantly reduces transponder counts across four WAN topologies compared to Shoofly-improved and FlexWAN at equivalent demand levels, achieving up to 91.9% average savings on the TWAN topology. This highlights its appeal for cloud service providers upgrading WANs. Flexnetic's transponder usage growth is nearly parallel yet lower than FlexWAN's, as its algorithm quickly identifies bottlenecked links and leverages SVTs' high capacity. For links with lower traffic, reusing existing BVT devices is more cost-effective. Unlike Shoofly-improved, which uses only BVTs and has an earlier termination line due to planning solver limitations, Flexnetic benefits from SVTs' higher capacity, accommodating greater traffic demands. Shoofly-improved fails to expand network capacity, underscoring Flexnetic's advantage of offering up to $2.33\times$ the capacity on B4.

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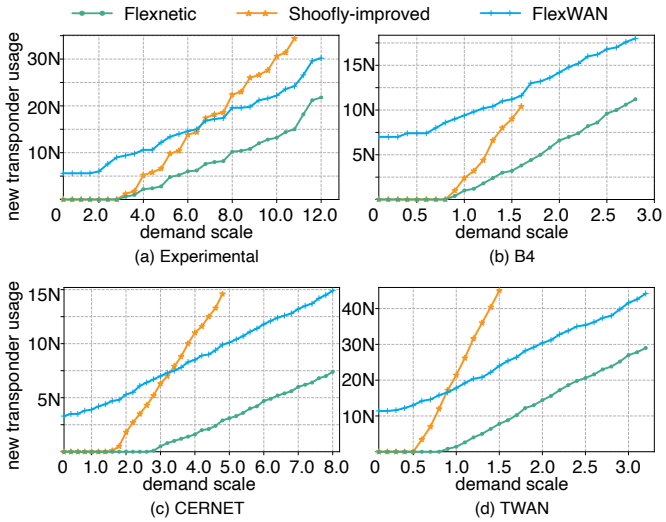

Fig. 5. Cost savings brought by Flexnetic.

B. Capacity of Unit Budget

SVT improves spectrum utilization by using advanced modulation formats, allowing a unit of spectrum to support higher IP link capacity [3]. Flexnetic repurposes existing BVTs, which carry considerable traffic on non-bottleneck links. Figure 6 shows the relationship between the number of new SVTs purchased for a network upgrade and the cumulative capacity of various IP links, indicating how upgrade costs correlate with the WAN's maximum capacity. It is clear that Flexnetic optimizes network upgrades by reusing BVTs. As demand scales, Flexnetic approaches the performance of FlexWAN, but generally remains the most cost-effective solution.

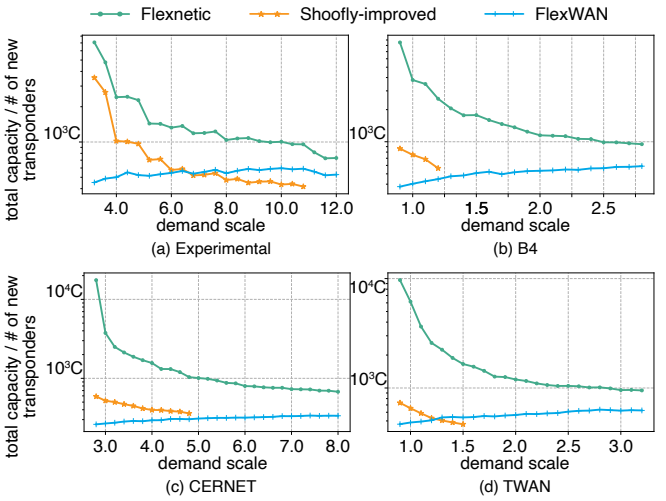

Fig. 6. Flexnetic offers a cost-effective WAN upgrade solution.

C. Solving Planning in Reality

The planning phase of Flexnetic utilizes MIP and is solved via Gurobi, a commercial solver. Since MIP problems are NP-hard, they present computational challenges. Our evaluation highlighted these difficulties, revealing that allowing for a gap in the solution process enhances practical problem-solving. Gurobi primarily employs the Branch and Bound method [9], optimizing its iterative steps with additional techniques. This method ensures optimality by aligning the feasible solution value with a theoretical lower bound. Achieving a zero gap indicates optimality, but reducing this gap, especially for medium-scale WAN topologies, can take extensive time (tens of hours). Our analysis shows Gurobi can attain a small non-zero gap quickly (within minutes). Figure 7 illustrates the gap-solving time relationship, confirming that tolerating a modest gap yields a "fairly good" solution swiftly. Flexnetic demonstrates significant SVT order savings with an acceptable 5% gap, which minimizes computational time without substantial trade-offs. This balance between computing time and SVT cost savings offers a strategic option for network operators.

We further examine the trade-off between solution quality and feasibility by analyzing Flexnetic's behavior under different solver gap thresholds. We verify that solutions obtained with a small gap (e.g., $\leq 5\%$) consistently satisfied all IP-layer capacity and optical-layer constraints, including spectrum continuity and modulation-reach feasibility. No infeasible mappings were observed. As such, setting the solver gap to 3–5% provides a practical trade-off, enabling fast planning without compromising correctness.

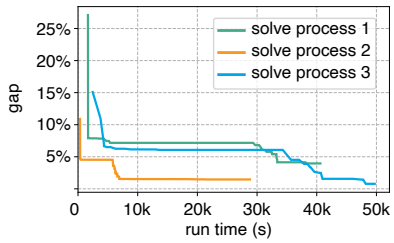

Fig. 7. Run time and gap during MIP processing.

VII. Related Work

Optical bypass. Optical bypass is crucial for improving transponder efficiency by avoiding unnecessary OEO conversions, as explored in studies like [2], [3], [10]–[13]. This paper contrasts the Shoofly project [2], which pioneered cost-saving methods in wide-area networks using optical bypass, though it faced certain challenges. Flexnetic enhances Shoofly's performance by addressing these issues with SVT devices and improved traffic prediction models.

Optical network planning. The network is hierarchical, with distinct rules at each level. Studies [1], [3], [10], [14]–[22] address optical technologies for robust WAN backbones. ARROW [10] focuses on recovering capacity during fiber failures with efficient online solutions. Flexnetic, which operates offline, examines WAN topology to plan upgrades and reduce capacity bottlenecks. FlexWAN [3], the first large-scale SVT device application in WAN planning, enhances network capacity using SVTs and introduces a complete optical layer constraint for these devices. Conversely, Flexnetic repurposes existing BVT devices, using SVTs as necessary, cutting costs compared to FlexWAN. Hose [15], developed by Meta, predicts inter-regional WAN traffic and reduces capacity needs via multiplexing, guiding phased network planning. RADWAN [1] suggests flexible WAN configurations with rate-variable optical devices, inspiring advanced planning technologies.

Traffic Engineering. Traffic engineering in WANs allows precise control over traffic direction and distribution, enhancing network management. Extensive research [8], [23]–[28]

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has contributed significantly to the design of the Flexnetic Capacity Predictor. Unlike traditional methods focused on static IP network traffic allocation, Flexnetic addresses capacity planning for IP Links based on traffic data.

VIII. Conclusion

In this paper, we introduce Flexnetic, a network upgrade planning tool designed primarily for cost reduction. Flexnetic integrates various approaches, including establishing optical bypasses, reusing existing optical equipment, and employing advanced spacing-variable optical hardware, to collectively save significant costs on unnecessary purchases of expensive optical devices for smooth WAN evolution. The solution process in Flexnetic is divided into two phases: capacity prediction and network planning, using an NLP solver and an MIP solver, respectively. We also explore potential issues in the optical bypass setup process and propose a heuristic method to mitigate the adverse effects of excessive optical bypasses on the network. Additionally, we investigate and trace the complex MIP problem-solving process in real-world scenarios, aiding operators in understanding the practical implications of optimal and suboptimal solutions. We simulated Flexnetic's performance in real WAN topologies, achieving up to 91.9% cost savings and $2.33\times$ capacity compared to previous work.

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References

[1] R. Singh, M. Ghobadi, K.-T. Foerster, M. Filer, and P. Gill, "Radwan: rate adaptive wide area network," in *Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication*, 2018, pp. 547–560.

[2] R. Singh, N. Bjorner, S. Shoham, Y. Yin, J. Arnold, and J. Gaudette, "Cost-effective capacity provisioning in wide area networks with shoofly," in *Proceedings of the 2021 ACM SIGCOMM 2021 Conference*, 2021, pp. 534–546.

[3] C. Miao, Z. Zhong, Y. Zhang, K. He, F. Li, M. Chen, Y. Zhao, X. Li, Z. He, X. Zou *et al.*, "Flexwan: Software hardware co-design for cost-effective and resilient optical backbones," in *Proceedings of the ACM SIGCOMM 2023 Conference*, 2023, pp. 319–332.

[4] IEEE, "Ieee p802.3ad link aggregation task force," 2000. [Online]. Available: https://www.ieee802.org/3/ad/

[5] Gurobi, "Gurobi optimization," 2023. [Online]. Available: https://www.gurobi.com

[6] H. Zang, J. P. Jue, B. Mukherjee *et al.*, "A review of routing and wavelength assignment approaches for wavelength-routed optical wdm networks," *Optical networks magazine*, vol. 1, no. 1, pp. 47–60, 2000.

[7] CERNET, "China education and research network," 2023. [Online]. Available: https://www.edu.cn/english/

[8] P. Kumar, Y. Yuan, C. Yu, N. Foster, R. Kleinberg, P. Lapukhov, C. L. Lim, and R. Soulé, "Semi-oblivious traffic engineering: The road not taken," in *15th USENIX Symposium on Networked Systems Design and Implementation (NSDI 18)*, 2018, pp. 157–170.

[9] D. R. Morrison, S. H. Jacobson, J. J. Sauppe, and E. C. Sewell, "Branch-and-bound algorithms: A survey of recent advances in searching, branching, and pruning," *Discrete Optimization*, vol. 19, pp. 79–102, 2016.

[10] Z. Zhong, M. Ghobadi, A. Khaddaj, J. Leach, Y. Xia, and Y. Zhang, "Arrow: restoration-aware traffic engineering," in *Proceedings of the 2021 ACM SIGCOMM 2021 Conference*, 2021, pp. 560–579.

[11] E. Modiano and A. Narula-Tam, "Mechanisms for providing optical bypass in wdm-based networks," *Optical Networks Magazine*, vol. 1, no. 1, pp. 11–18, 2000.

[12] J. M. Simmons and A. A. Saleh, "The value of optical bypass in reducing router size in gigabit networks," in *1999 IEEE International Conference on Communications (Cat. No. 99CH36311)*, vol. 1. IEEE, 1999, pp. 591–596.

[13] M. Chamania, M. Caria, and A. Jukan, "Achieving ip routing stability with optical bypass," *Optical Switching and Networking*, vol. 7, no. 4, pp. 173–184, 2010.

[14] Y. Xia, Y. Zhang, Z. Zhong, G. Yan, C. L. Lim, S. S. Ahuja, S. Bali, A. Nikolaidis, K. Ghobadi, and M. Ghobadi, "A social network under social distancing: Risk-driven backbone management during covid-19 and beyond," in *18th USENIX Symposium on Networked Systems Design and Implementation (NSDI 21)*, 2021, pp. 217–231.

[15] S. S. Ahuja, V. Gupta, V. Dangui, S. Bali, A. Gopalan, H. Zhong, P. Lapukhov, Y. Xia, and Y. Zhang, "Capacity-efficient and uncertainty-resilient backbone network planning with hose," in *Proceedings of the 2021 ACM SIGCOMM 2021 Conference*, 2021, pp. 547–559.

[16] A. K. Bangla, A. Ghaffarkhah, B. Preskill, B. Koley, C. Albrecht, E. Danna, J. Jiang, and X. Zhao, "Capacity planning for the google backbone network," 2015.

[17] J. Gossels, G. Choudhury, and J. Rexford, "Robust network design for ip/optical backbones," *Journal of Optical Communications and Networking*, vol. 11, no. 8, pp. 478–490, 2019.

[18] F. S. Abkenar and A. G. Rahbar, "Study and analysis of routing and spectrum allocation (rsa) and routing, modulation and spectrum allocation (rmsa) algorithms in elastic optical networks (eons)," *Optical Switching and Networking*, vol. 23, pp. 5–39, 2017.

[19] M. Birk, G. Choudhury, B. Cortez, A. Goddard, N. Padi, A. Raghuram, K. Tse, S. Tse, A. Wallace, and K. Xi, "Evolving to an sdn-enabled isp backbone: Key technologies and applications," *IEEE Communications Magazine*, vol. 54, no. 10, pp. 129–135, 2016.

[20] A. L. Chiu, G. Choudhury, G. Clapp, R. Doverspike, M. Feuer, J. W. Gannett, J. Jackel, G. T. Kim, J. G. Klincewicz, T. J. Kwon *et al.*, "Architectures and protocols for capacity efficient, highly dynamic and highly resilient core networks," *Journal of Optical Communications and Networking*, vol. 4, no. 1, pp. 1–14, 2011.

[21] V. Dukic, G. Khanna, C. Gkantsidis, T. Karagiannis, F. Parmigiani, A. Singla, M. Filer, J. L. Cox, A. Ptasznik, N. Harland *et al.*, "Beyond the mega-data center: networking multi-data center regions," in *Proceedings of the Annual conference of the ACM Special Interest Group on Data Communication on the applications, technologies, architectures, and protocols for computer communication*, 2020, pp. 765–781.

[22] X. Jin, Y. Li, D. Wei, S. Li, J. Gao, L. Xu, G. Li, W. Xu, and J. Rexford, "Optimizing bulk transfers with software-defined optical wan," in *Proceedings of the 2016 ACM SIGCOMM Conference*, 2016, pp. 87–100.

[23] J. Bogle, N. Bhatia, M. Ghobadi, I. Menache, N. Bjørner, A. Valadarsky, and M. Schapira, "Teavar: striking the right utilization-availability balance in wan traffic engineering," in *Proceedings of the ACM Special Interest Group on Data Communication*, 2019, pp. 29–43.

[24] H. H. Liu, S. Kandula, R. Mahajan, M. Zhang, and D. Gelernter, "Traffic engineering with forward fault correction," in *Proceedings of the 2014 ACM Conference on SIGCOMM*, 2014, pp. 527–538.

[25] F. Abuzaid, S. Kandula, B. Arzani, I. Menache, M. Zaharia, and P. Bailis, "Contracting wide-area network topologies to solve flow problems quickly," in *18th USENIX Symposium on Networked Systems Design and Implementation (NSDI 21)*, 2021, pp. 175–200.

[26] C.-Y. Hong, S. Mandal, M. Al-Fares, M. Zhu, R. Alimi, C. Bhagat, S. Jain, J. Kaimal, S. Liang, K. Mendelev *et al.*, "B4 and after: managing hierarchy, partitioning, and asymmetry for availability and scale in google's software-defined wan," in *Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication*, 2018, pp. 74–87.

[27] S. Jain, A. Kumar, S. Mandal, J. Ong, L. Poutievski, A. Singh, S. Venkata, J. Wanderer, J. Zhou, M. Zhu *et al.*, "B4: Experience with a globally-deployed software defined wan," *ACM SIGCOMM Computer Communication Review*, vol. 43, no. 4, pp. 3–14, 2013.

[28] C.-Y. Hong, S. Kandula, R. Mahajan, M. Zhang, V. Gill, M. Nanduri, and R. Wattenhofer, "Achieving high utilization with software-driven wan," in *Proceedings of the ACM SIGCOMM 2013 Conference on SIGCOMM*, 2013, pp. 15–26.

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